ANNEXURE 2

Budget & Bill of Materials

Component-wise cost estimation for the SmartFire Rover prototype.

Currency	BDT (Tk.) — Bangladeshi Taka; USD estimate also shown
Sourcing	Local electronics shops (Stadium Market, Mirpur-10)
Quote validity	April 2026

Hardware Components

No.	Component	Qty	Unit (Tk.)	Total (Tk.)
1	Arduino Uno R3 (clone)	1	600	600
2	IR Flame Sensor Module	3	90	270
3	L298N Dual H-Bridge Motor Driver	1	140	140
4	5V Single Channel Relay Module	1	90	90
5	LM2596 DC-DC Buck Regulator	1	130	130
6	SG90 Micro Servo Motor	1	140	140
7	12V Mini DC Water Pump	1	200	200
8	Silicone Water Tube + Nozzle	1	50	50
9	NEO-6M GPS Module	1	500	500
10	ESP8266 (ESP-01) Wi-Fi Module	1	300	300
11	4WD Robot Car Chassis Kit	1	1020	1,020
12	18650 Li-ion Cell 3.7V	2	150	300
13	18650 Battery Holder (2-cell)	1	120	120
14	Jumper Wires	1	150	150
15	PVC Board	1	2000	200
16	Glues, Tapes	--	--	120
	GRAND TOTAL			**Tk. 4,350**

Cost Breakdown by Subsystem

Subsystem	Cost (Tk.)	% of build	Items
Microcontroller & logic	750	11.9%	Arduino Uno
Sensing	420	6.6%	IR x3
Communications & GPS	1,070	16.9%	ESP8266 + NEO-6M
Actuation	970	15.3%	Pump + servo + relay + tubing
Drive train & power	2,330	36.9%	Chassis + motors + battery + LM2596
Misc. & assembly	780	12.3%	Wiring, mounts, consumables

Notes

• Prices are local market quotes (Stadium Market, Dhaka) and online listings; exact bill may vary.

• Chassis kit includes 4x yellow gear motors and wheels.

• Tools (soldering iron, multimeter, USB cable) assumed available in lab — not included in budget.

• Software stack (Arduino IDE, Tinkercad, libraries) is free / open-source.